

Session 08 - Q & A

Questions that we should be able to answer

- We need to be able to draw architecture (about 3 different ones for a given problem in the exam)
 - Since we can draw anything they also ask: "Please explain why you draw the model like you did?"
 - What is the data flow through the model?
 - What is the motivation behind the model you made?
 - Last year 7 total Tasks in the exam
 - No TensorFlow code is required, but you might need to **describe** how a model works or is structured.
 - Focus on **intuition, structure, and application** of concepts discussed in class and exercises.
 - Practice exercises are the best preparation. Focus on model types like CNNs, RNNs, LSTMs, GANs, and Transformers.
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- **? What is unsupervised learning?**
 - Be able to explain what it means, what types of problems it solves, and give an example (e.g., clustering or dimensionality reduction).
 - **? What is regression in machine learning?**
 - Explain how it differs from classification, and give an example of a regression task (e.g., predicting temperature).
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- **? What kinds of problems are Convolutional Neural Networks (CNNs) particularly good at solving? Why?**
 - Think image data, spatial hierarchies, feature extraction.
 - **? What types of layers are typically used in a CNN architecture?**
 - Convolutional layers, pooling layers, activation layers (e.g., ReLU), and fully connected (dense) layers.
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- **? You are given accelerometer data. How would you design a neural network to detect walking vs. running?**
 - What is the input shape?
 - What kind of layers would you use (e.g., RNN, LSTM, CNN)?
 - What does the output look like?
 - **? Design a neural network for a time-series classification problem. Draw the architecture and describe the role of each component.**
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- **? What is important when creating a dataset for a supervised learning task?**
 - Class balance, preprocessing, and **splitting data** into training/validation/test sets.
- **? What are a generator and discriminator in a GAN, and how do they interact?**
 - Adversarial training loop.
 - Freezing and deployment
- **? How would you design a system that generates images of fruits using a GAN?**

- **? How does a transformer-based model generate language?**
 - Know the basic mechanism (attention, tokens, decoder), but no deep implementation details.
- **? What is online ML and how does it differ from ML?**
 - Look at the chapter about online ML
- **? How could a model engage in a positive, adaptive loop with a human (e.g., in a game or assistive setting)?**

Example of how a architecture that we need to draw might look like:

